Unit 2 - Analysing Epidemiological Data

Prevelance - Proportion / % of population who have specific characteristics @ specific point in time.

$$\text{prevelance} = \frac{\text{\# of people in sample with characteristic}}{\text{total \# people in sample}}$$

prevelance percentage $\times 100$

Difficult to determine covid19 prevelance:

How many people currently infected?
- When test positive?
- When infected?
- How long infected?
- How many cases not detected?

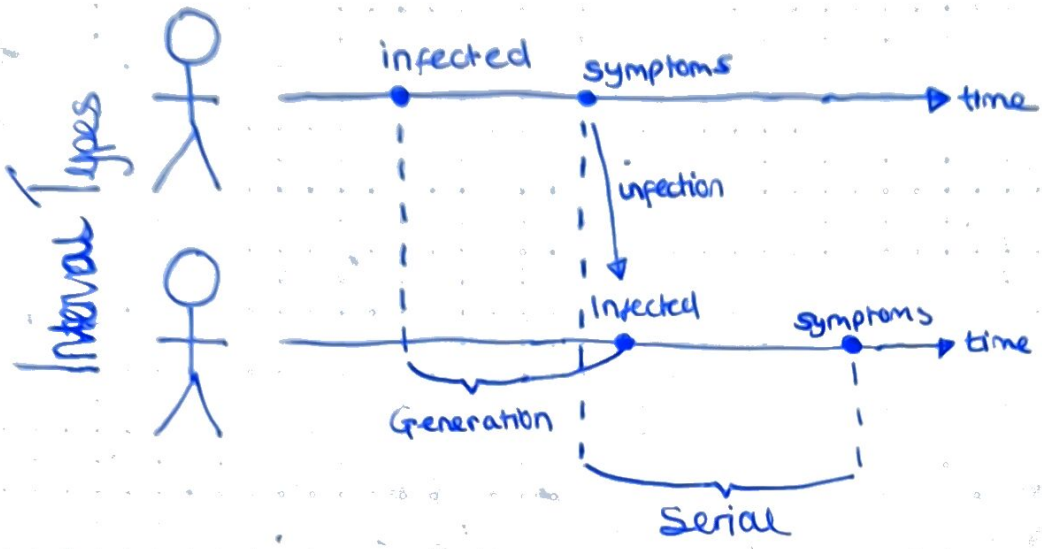
Incidence - # new instances of specific characteristic/event that occur in a population over specified time period

$$\text{Incidence} = \frac{(\text{\# new cases/events in time frame}) \times (\text{desired time frame})}{(\text{total \# people in population}) \times (\text{observed time frame})}$$

$= _$ per x population $\times$ x

Detected Incidence not actual

Incubation period - time from exposure to a disease to presentation of symptoms



Case fatality rate - proportion of cases resulting in death
Epidemic - # cases of infectious disease in excess of expected frequency
Host - Living species (person) capable of being infected
Risk - Probability that event/outcome/disease/condition will develope in time frame
Sensitivity - Probability of individual with disease will get positive test result
Specificity - Probability of individual without diseases will get negative test result
Surveillance - Collection of data on a particular disease
Susceptible - An idividual at risk of contracting disease

Exponential Growth - Rate of change proportional to current quantity

Growth Rate

$$I(n+1) = \lambda I(n)$$

(The following day)

$I(n)$ = # infectious people on day n

λ = daily growth rate

If $\lambda > 0$ then # infectious increase

If $\lambda < 0$ then # of infections decrease

If $\lambda = 0$ then constant

More exact estimate of growth rate:

$$100 \times (e^{\lambda} - 1)\%$$

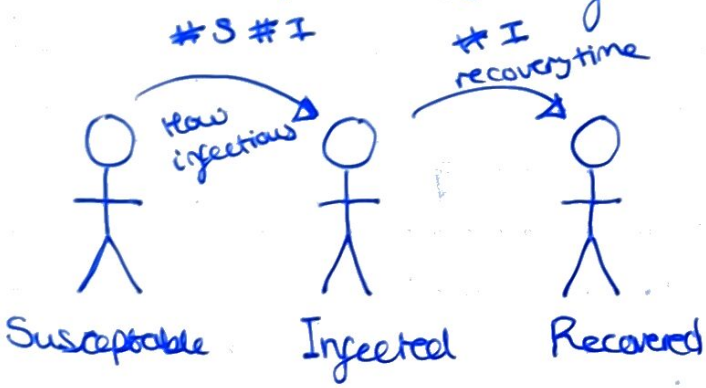
exponential

Doubling Time = Time taken for cases to double

When decreasing known as half life

% > % = pipe

table_name $ column → gets column from tibble (table)



Basic Reproduction # = expected # of new cases directly generated by 1 case in a population (all individuals are susceptable to infection)

This value is called R_0

Effective reproduction # = Expected # new cases directly generated by 1 case in current population

This value is called R_e

Estimate R_0 using data from start of an outbreak
Estimate R_e using most recent data

If $R_0 > 1$ = outbreak
If $R_0 < 1$ = Extinction

R_0 = how likely Growth Rate = how quickly

Models are only as good as assumptions

Serial interval - time Spent infectious

New assumption = Shorter but more confident